

$\text{path} = \{ \cancel{a}, \cancel{a}, \cancel{b}, \cancel{b}, \cancel{bb}, \cancel{aa}, \cancel{b}, \cancel{b}, \cancel{bb} \}$
 $\text{res} = \{ \{a, a, b, b\}, \{a, a, bb\}, \{aa, b, b\}, \{aa, bb\} \}$

~~aaabbb~~

$a \ a \ b \ b$, index=0
 $i = \cancel{0} \ \cancel{1} \ \cancel{2} \ 3 \ 4$

$a \ b \ b$, index=1
 $i = \cancel{1} \ \cancel{2} \ 3 \ 4$

$b \ b$, index=2
 $i = \cancel{2} \ \cancel{3} \ 4$

$b \ b$, index=2
 $i = \cancel{2} \ \cancel{3} \ 4$

b , index=3
 $i = \cancel{3} \ 4$
 index=4

index=4
 b , index=3
 $i = \cancel{3} \ 4$
 index=4

$n = 4 = \text{length of string}$

a a b b

0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

No. of ways

we can do

partitions $= 2^{n-1}$

After doing partition
for each part

we check whether
it is a palindrome
or not. So, for
each part when we
combine time
complexity $T = O(n)$

$$S, T = O(n) * O(2^{n-1}) \\ = O(n \cdot 2^n)$$

$$S = O(n) + O(n) = O(n)$$

Stack can
grow max
of n height

path can be
max $O(n)$